

SOPHYANE

Complete Product Handbook

Version 17.1.0 | Cross-Platform Local Agentic AI Harness

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Repository: github.com/badrpk/sophyane

Public site / API: xerus.biz (portal :8780)

One stack for agents on phone, PC, cloud, industrial gateways, and Linux SoCs.

Harness · Mesh · Federated PEFT · Ultra-low API · Appliance boot · ERP/Apps

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1. Executive Summary

Sophyane 17.1.0 is an open agent infrastructure layer: plan-act-verify harness, multi-provider LLM routing with automatic open-model rescue, multi-agent runtime, device mesh, C++ continual PEFT training, SoC appliance boot, hardware multi-language API, investor cloud portal with sph_ API keys, and ultra-low hybrid pricing that pushes heavy compute onto user-owned hardware.

This handbook is the complete operator and product reference for install, features, APIs, hosting, competitive position, and security.

2. Product Definition

Is

- Agent OS / harness for developers and edge fleets
- Local-first with cloud API fallback
- Federated learning + mesh compute network
- Public token API with extreme low prices
- Installable OS-like appliance on Linux SoCs

Is not

- Bare-metal MCU firmware without Linux
- A full Cursor IDE replacement alone
- Silent compute theft (train is opt-in)

3. Architecture

```
Clients: CLI/TUI | Cloud portal | C++/JS/Python SDKs | Mesh peers
|
Control: v13_cli, skills, RAG, HITL, budget, permissions, traces
|
Harness: plan -> tool act -> observe -> verify -> repair
|
Providers: OpenAI/Gemini/xAI/Ollama/local_gguf + FallbackProvider
|
Kernel bus: hardware | mesh | erp | app_factory | agents | memory
|
Services: :8766 llama | :8770 HW API | :8777 mesh | :8780 cloud portal
|
Train: sophyane-train-core (C++ PEFT + FedAvg) over GGUF base_hash
```

4. Install & Upgrade

Public installer always selects the newest release/vX.Y.Z or tag from GitHub, verifies commit, builds venv, installs package, writes ~/.local/bin/sophyane, optional C++ train core.

```
curl -fsSL https://raw.githubusercontent.com/badrp/sophyane/main/install.sh | sh
sophyane --version && sophyane --doctor && sophyane --audit && sophyane --capabilities
```

Pin version: SOPHYANE_VERSION=17.1.0 curl ... | sh. Pin commit: SOPHYANE_COMMIT=<sha>.

5. CLI / TUI

Run sophyane with no args for Grok-style interactive mode. Slash commands: /help /status /model /local /doctor /new /quit /session-info.

6. Providers & Local Models

Plugins cover major clouds plus Ollama and local_gguf (llama-server default :8766). Fallback detects quota/credit/auth errors and bootstraps RAM-fit GGUF. Hardware tiers nano/micro/small/standard.

7. Agent Harness

Strict JSON plans, allowlisted tools, path jail, deterministic verification, iteration caps, nested timeouts, full traces. Prefer plan-and-execute for long coding tasks.

8. Coding Runtime

RepositoryIndex symbols/imports/tests/digests; apply_patch; pytest-driven self-repair; refuse test deletion greenwashing; execution evidence with SHA-256.

9. Skills · RAG · REPL · MCP-lite

Feature	Command	Notes
Skills	--skills --skill NAME	Builtin + ~/.local/share/sophyane/skills
RAG	--rag-add --rag-query	Local JSONL TF-IDF-ish
REPL	--repl CODE	AST-blocked os/subprocess
MCP-lite	--mcp-list --mcp-call	JSON tools over HTTP/CLI

10. Multi-Agent · Graph · Daemon · Scheduler

- Supervisor/worker multi-agent with unique threads and lifecycle persistence
- graph_runtime state graphs (LangGraph-comparable core without hard dep)
- Daemon queue ticks; scheduler jobs --schedule / --schedule-run

11. Safety Stack

Control	CLI	Purpose
Guardrails	built-in	Block destructive shell patterns
HITL	--hitl-request --approve	Human approval queue
Budget	--budget-status	Token/cost caps
Permissions	--permissions PROFILE	readonly/workspace/network/full

12. Memory · Checkpoints · Observability

- SQLite memory store for sessions
- Checkpoints --checkpoint-list for long tasks
- Traces --trace-list JSONL spans
- Notifications log / notify-send / webhook

13. Mesh Federation

Port 8777. Discover LAN/USB/ADB, clone install, shared storage, compute offload. Hello magic SOPHYANE_MESH_v1. SOPHYANE_MESH_TOKEN optional. Boot reuses healthy listeners (idempotent).

```
sophyane --mesh-serve
sophyane --mesh-discover
sophyane --mesh-install HOST --yes
sophyane --mesh-compute "summarize status"
```

14. Continual Federated Training

C++ core local-step + FedAvg. Base GGUF stays local; only adapter deltas share. Opt-in. Digests by default. Goal: improve shared Sophyane adapters across millions of installs.

```
sophyane --train-build-core
sophyane --train-opt-in
sophyane --train-step
sophyane --train-round
sophyane --train-status
```

15. AI Kernel · ERP · App Factory

Userspace kernel modules coordinate hardware/software/apps/ERP/mesh/agents. ERP: Oracle SAP Odoo Dynamics NetSuite ERPNext. Apps: web Android Harmony iOS desktop API scaffolds.

16. Hardware API & Vendors

25+ chip vendors catalogued. HTTP API :8770. SDKs in sdk/cpp, sdk/js, sdk/python. Routes: /v1/hardware/*, /v1/kernel, /v1/erp, /v1/apps/create, /v1/train.

17. Appliance Boot

sophyane --boot: platform, Ethernet/Wi-Fi bring-up, kernel, HW API, mesh, improve heartbeat. --install-chip / --install-appliance-unit for systemd user unit. Not a silicon bootloader.

18. Browser · Web Intel · Self-Improve

- sophyane-browser Chromium profile + home UI
- --fetch / --learn for scrape + ledger blocks
- Daily improve export + GitHub Action workflow

19. Cloud Portal · Tokens · Pricing

Website + API on :8780. Signup returns sph_ keys (shown once). Plans designed for mass adoption; hybrid free edge compute on user devices.

Plan	USD/mo	Included tokens	Overage /1M
free	0	500,000	0.05
hybrid	0	2,000,000 + free edge	0.02
builder	1	10,000,000	0.03
scale	9	200,000,000	0.015

```
POST /api/v1/signup {email, plan}
```

```
POST /api/v1/chat Authorization: Bearer sph_... {message, edge?}
```

```
GET /api/v1/usage | /api/v1/pricing | /api/v1/health
```

20. DNS · Hosting · Tunnel

Namecheap API user badrpk. Domain xerus.biz pointed for Sophyane. Static IP 154.57.212.38. On Crostini, inbound :80 may be closed; use Cloudflare tunnel and/or URL records so browsers reach the portal.

```
sophyane --namecheap-domains
```

```
sophyane --namecheap-setup-site --static-ipv4 154.57.212.38
```

```
sophyane --cloud-serve --cloud-port 8780
```

21. Competitive Exam (v17)

Live: tough100 100/100, harness PASS, multi-agent PASS, evidence PASS, capabilities ~96%, cloud healthy. Matrix vs 14 ecosystems: Sophyane rank #1 average ~8.5 on infrastructure dimensions. Cursor wins IDE polish; Replit wins hosting UX; LangGraph wins visual graph ecosystem.

22. Full CLI Map (selected)

Flag / command	Purpose
--doctor	Diagnostics
--audit	Integrated feature audit
--capabilities	Capability matrix
--platform	Host probe
--edge-health	Edge profile
--hardware	Vendor/HW report
--hardware-api	Serve :8770
--kernel	Boot AI kernel
--create-app	Scaffold app
--erp	ERP status
--mesh-serve	Mesh :8777
--mesh-discover	Find peers
--mesh-compute	Remote compute
--boot	Appliance boot
--install-chip	Chip install script
--train-opt-in	Allow PEFT contrib
--train-step	Local C++ step
--train-round	Fed round
--cloud-serve	Website+API :8780
--skills	List skills
--rag-add/--rag-query	Local RAG
--repl	Sandboxed Python
--mcp-list/--mcp-call	MCP-lite
--schedule*	Cron jobs
--budget-status	Token budgets
--hitl-*	Approvals
--trace-list	Observability
--exam-tough100	100-question exam
--ask	Expert hybrid Q&A
--namecheap-*	DNS automation
--browser	Sophyane browser
--fetch/--learn	Web intel

23. Paths & Config

Path	Role
~/config/sophyane/	llm.json, secrets, namecheap.env
~/local/share/sophyane/	releases, current, venv, models, website
~/local/state/sophyane/	portal.db, traces, train, mesh, rag
~/local/bin/sophyane	CLI entry

sdk/cpp/continual/

C++ train core sources

24. Runbooks

Start full local stack

```
sophyane --doctor
sophyane --cloud-serve --cloud-host 127.0.0.1 --cloud-port 8780 &
# nginx proxy xerus.biz -> 127.0.0.1:8780
# optional cloudflared tunnel --url http://127.0.0.1:80
```

Issue API key

```
curl -s http://127.0.0.1:8780/api/v1/signup -H 'Content-Type: application/json' \
-d '{"email":"you@domain.com","plan":"hybrid"}'
```

Backup

- Copy ~/.config/sophyane and ~/.local/state/sophyane regularly

25. Security

- chmod 600 all env files with keys
- Rotate Namecheap/API passwords if ever pasted in chat
- Mesh token on untrusted LANs
- Train opt-in; digests default
- REPL is not a full VM sandbox
- Terminate public portal with TLS in production
- Do not commit portal.db or secrets to git

26. Roadmap Honesty

- Stable named Cloudflare tunnel or real public :443
- Stripe billing beyond usage counters
- Full duplex voice + full computer-use
- Stronger local coding models via PEFT distillation
- Official mobile companion apps

27. Glossary

- GGUF: llama.cpp model package format
- PEFT/LoRA: Small adapters instead of full fine-tune
- FedAvg: Average adapters across peers
- HITL: Human approval gate
- Mesh: Device federation network
- Harness: Verified agent tool loop
- sph_: Cloud API key prefix
- Appliance: Auto-boot agent services on Linux SoC

28. Appendix: Quick Start Scripts

```
# Fresh machine
curl -fsSL https://raw.githubusercontent.com/badrpk/sophyane/main/install.sh | sh
sophyane --setup # providers
sophyane --local # open model if needed
sophyane --boot
sophyane --cloud-serve &
sophyane --train-opt-in && sophyane --train-step
```

End of Sophyane 17.1.0 Complete Handbook.

Repo <https://github.com/badrpk/sophyane>

Handbook generator: [scripts/generate_handbook_pdf.py](#)